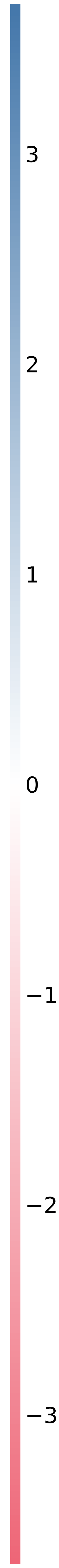
Question:
A 7-year-old boy presents to your office with facial eczema. He has a history of recurrent infections, including multiple episodes of pneumonia that lasted several weeks and otitis media. Laboratory measurements of serum immunoglobulins show increased IgE and IgA but decreased IgM. Which of the following additional abnormalities would you expect to observe in this patient?
Answer Choices:
A. Thrombocytopenia
B. Leukopenia
C. Anemia
D. NADPH oxidase deficiency

Reasoning + Answer:



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The patient is a 7-year-old boy with facial eczema, recurrent infections (pneumonia, otitis media), elevated IgE and IgA, and decreased IgM. This clinical picture is highly suggestive of selective IgM deficiency, which is often associated with a broader immunodeficiency phenotype. Importantly, recurrent bacterial infections, especially respiratory ones, and elevated IgE are characteristic of a disorder involving B-cell dysfunction.

The key clue is the normal or elevated IgE and IgA with decreased IgM. This pattern is commonly seen in patients with **hyper-IgE syndrome**, particularly the most common form known as **autosomal dominant hyper-IgE syndrome (HIES)**. This condition is caused by mutations in the *STAT3* gene, which affects interleukin-6 (IL-6) signaling and downstream cytokine responses.

In classic HIES:
- Marked elevation of serum IgE (often > 2000 IU/mL)
- Normal or elevated IgA
- Decreased IgM
- Recurrent skin infections (eczema), sinusitis, pneumonia
- Frequent staphylococcal skin and lung infections
- Characteristic facial eczema and enlarged nasal passages

A hallmark of HIES is **thrombocytopenia**, which is often observed due to the underlying immune dysregulation and bone marrow involvement. The thrombocytopenia is not due to a defect in platelet production per se but is a consequence of immune dysregulation and inflammation.

Now, evaluating the options:
A. Thrombocytopenia âĢĵ YES, commonly seen in HIES.
B. Leukopenia âĢĵ not typical; leukocytosis is more common.
C. Anemia âĢĵ not a consistent feature.
D. NADPH oxidase deficiency âĢĵ this is seen in chronic granulomatous disease (CGD), which presents with recurrent pyogenic infections and neutrophil dysfunction, but not with elevated IgE or eczema. CGD does not have the IgE pattern.

Thus, the most likely additional abnormality in this context is thrombocytopenia.

<answer>
A. Thrombocytopenia
</answer>